

# Public Health Analytics and Early Warning Platform

## Technical White Paper for AI-Assisted Syndromic Surveillance Modernization

Prepared as an independent public health analytics research prototype. This document describes a modern surveillance analytics architecture for transforming public aggregate or synthetic emergency department trend data into explainable signal intelligence, geographic risk awareness, and analyst-facing decision support.

| Document Type | Technical white paper                                                                                                           |
|---------------|---------------------------------------------------------------------------------------------------------------------------------|
| System Focus  | Syndromic surveillance analytics, anomaly detection, forecasting, geospatial risk, RAG/NLP search, and synthetic HL7 simulation |
| Data Boundary | Public aggregate/synthetic surveillance data; 50,000 raw rows and 182,241 processed analytic records                            |
| Prepared Date | May 11, 2026                                                                                                                    |

## Executive Summary

Public health surveillance is a core component of health security because it enables agencies to detect unusual illness patterns, monitor emerging threats, and support preparedness before signals become operationally severe. Syndromic surveillance is especially important because it uses timely healthcare encounter patterns, such as emergency department visits and symptom groupings, before confirmed diagnoses are available.

The Public Health Analytics and Early Warning Platform demonstrates how modern data engineering, transparent statistical monitoring, machine learning anomaly detection, forecasting, geographic risk scoring, and retrieval-augmented search can be integrated into a single analyst-facing surveillance workflow. The system converts public aggregate or synthetic emergency department trend data into structured outputs that support review, triage, communication, and response planning.

The platform is not designed to replace epidemiological judgment or official public health decision-making. Its value is in the engineering pattern: a modular pipeline that ingests surveillance data, standardizes it, computes temporal features, detects unusual patterns, scores risk, displays geographic context, summarizes data quality, and allows analysts to ask grounded natural-language questions about the generated outputs.

| Metric                     | Current Value            |
|----------------------------|--------------------------|
| Raw source rows            | 50,000                   |
| Processed analytic records | 182,241                  |
| Date range                 | 2022-10-01 to 2026-04-30 |
| States / regions           | 51                       |
| Syndrome categories        | 11                       |
| Anomaly flags              | 5,468                    |
| High/Critical records      | 2,396                    |
| Data completeness          | 99.58%                   |

## Public Health Context

Syndromic surveillance supports early detection, situational awareness, and monitoring of health threats by tracking symptoms and visit patterns before final diagnosis confirmation. This approach is relevant for respiratory illness, outbreak monitoring, opioid and overdose surveillance concepts, heat illness, environmental exposures, disaster response, and emergency preparedness.

The system design is conceptually aligned with public health data modernization themes reflected in CDC NSSP, ESSENCE, the BioSense Platform, HHS preparedness and data modernization priorities, and WHO epidemic intelligence practices. These systems and initiatives emphasize timeliness, interoperability, cross-jurisdiction awareness, data quality, and actionable public health intelligence.

| Public Health Need           | Platform Response                                                                                     |
|------------------------------|-------------------------------------------------------------------------------------------------------|
| Earlier signal review        | Rolling baselines, z-score anomaly detection, Isolation Forest scoring, and risk-ranked alert tables. |
| Cross-jurisdiction awareness | State-level geospatial risk aggregation, rankings, and heat-map style visualization.                  |
| Data modernization           | Reusable ETL, documented data dictionary, model summary, and modular source code.                     |
| Preparedness analytics       | Forecasting, signal summaries, policy briefing prompts, and data quality indicators.                  |
| Interoperability readiness   | Synthetic HL7 generation concepts and future HL7/FHIR ingestion roadmap.                              |

## Reference Architecture

The platform is organized as a modular surveillance analytics architecture. Each layer has a distinct role so the system can be tested, extended, governed, and eventually deployed in more scalable environments.

| Layer                   | Implementation Role                                                                                                                                                  |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data ingestion          | Loads public/synthetic surveillance extracts and cached processed datasets. Future versions can ingest API feeds, HL7 messages, FHIR resources, or streaming events. |
| ETL and quality control | Standardizes dates, geography, syndrome labels, numeric values, duplicates, missingness, and analytic grain.                                                         |
| Feature engineering     | Computes lag features, rolling baselines, rolling standard deviation, z-scores, percent change, trend acceleration, and persistence indicators.                      |
| Analytics engine        | Runs statistical anomaly detection, Isolation Forest anomaly scoring, short-term forecasting, and model output generation.                                           |
| Risk scoring            | Combines severity, growth, anomaly status, and persistence into a bounded score and categorical risk level.                                                          |
| Geospatial intelligence | Aggregates risk by state/region and supports risk ranking and map-based triage.                                                                                      |
| RAG/NLP assistant       | Retrieves local documentation and live surveillance summaries to answer natural-language analyst questions.                                                          |
| Dashboard/API layer     | Presents filters, KPIs, charts, alert tables, methods, evidence, and future API-ready outputs.                                                                       |

## Cloud-Native Scaling Path

A production-oriented implementation could evolve toward an event-driven cloud architecture. Batch files or API extracts could land in object storage, validation jobs could enforce schema and quality rules, feature pipelines could publish curated analytic tables, model scoring could run on a schedule or trigger, and dashboard/API services could present only approved aggregate outputs. Streaming ingestion through managed event buses or Kafka-style systems would improve timeliness but would require schema governance, idempotency controls, backpressure handling, observability, and security controls.

## Data Pipeline and Current Dataset

The current processed dataset contains state-level syndrome time series derived from public aggregate or synthetic emergency department visit trajectory data. The processing workflow reshapes wide syndrome percentage fields into a long analytic table and computes features by state and syndrome.

| Data Product        | Value                                                 |
|---------------------|-------------------------------------------------------|
| Raw file            | data/raw/cdc_nssp_ed_trajectories_raw.csv             |
| Processed file      | data/processed/cdc_nssp_ed_trajectories_processed.csv |
| Processed records   | 182,241                                               |
| Syndrome categories | 11                                                    |
| Missing-value rate  | 0.42%                                                 |
| Duplicate records   | 0                                                     |

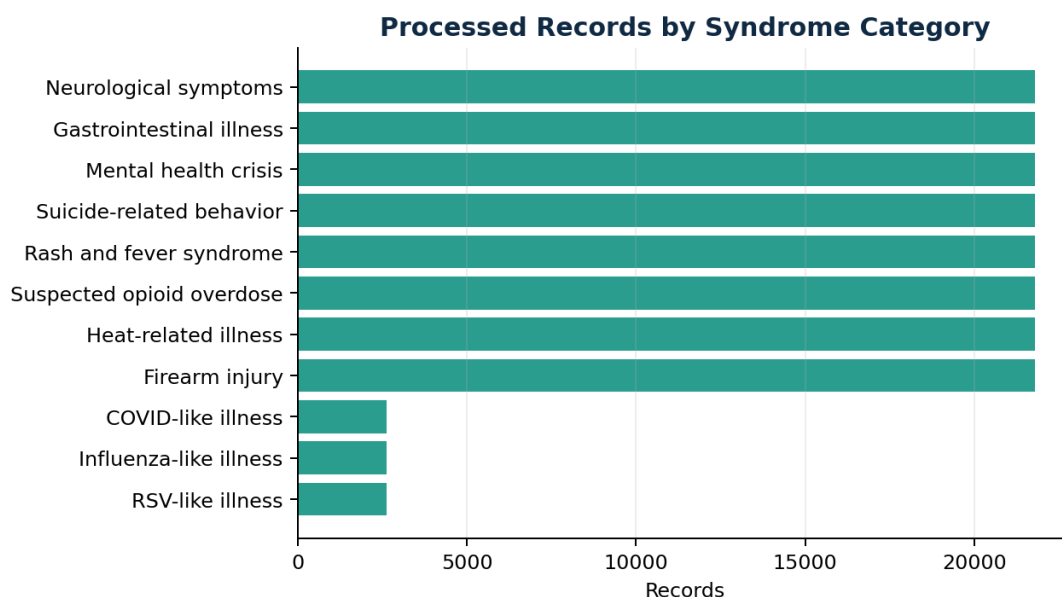


Figure 1. Distribution of processed analytic records by syndrome category in the current dataset.

## AI/ML Methodology

### Rolling Baseline and Statistical Anomaly Detection

The first model layer uses rolling baseline comparison. For each state and syndrome category, the system computes a rolling mean and rolling standard deviation, then calculates a z-score for each observation. This provides an interpretable measure of how unusual current activity is relative to recent local behavior.

### Isolation Forest Anomaly Detection

The second model layer uses Isolation Forest, an unsupervised anomaly detection method suitable for settings where labeled outbreak data is limited. The model evaluates multivariate feature patterns including observed visit percentage, rolling baseline, z-score, percent change, and trend acceleration. This provides a complementary signal to threshold-based statistical detection.

### Composite Risk Scoring

The risk engine combines statistical severity, positive recent growth, Isolation Forest anomaly status, and persistence of recent anomaly activity. The result is a 0-100 risk score mapped to Low, Moderate, High, or Critical. This design supports prioritization and triage while keeping outputs reviewable by analysts.

| Model Component  | Purpose                                          | Operational Caveat                                               |
|------------------|--------------------------------------------------|------------------------------------------------------------------|
| Rolling baseline | Local expected value for state/syndrome behavior | Can be unstable at the start of a time series.                   |
| Z-score          | Transparent standardized deviation from baseline | Requires stable rolling variance.                                |
| Isolation Forest | Unsupervised multivariate anomaly detection      | Requires calibration and analyst review.                         |
| Percent change   | Captures recent growth against lagged values     | Can be volatile when prior values are small.                     |
| Risk score       | Combines multiple signals into review priority   | Weights should be governed and validated before operational use. |

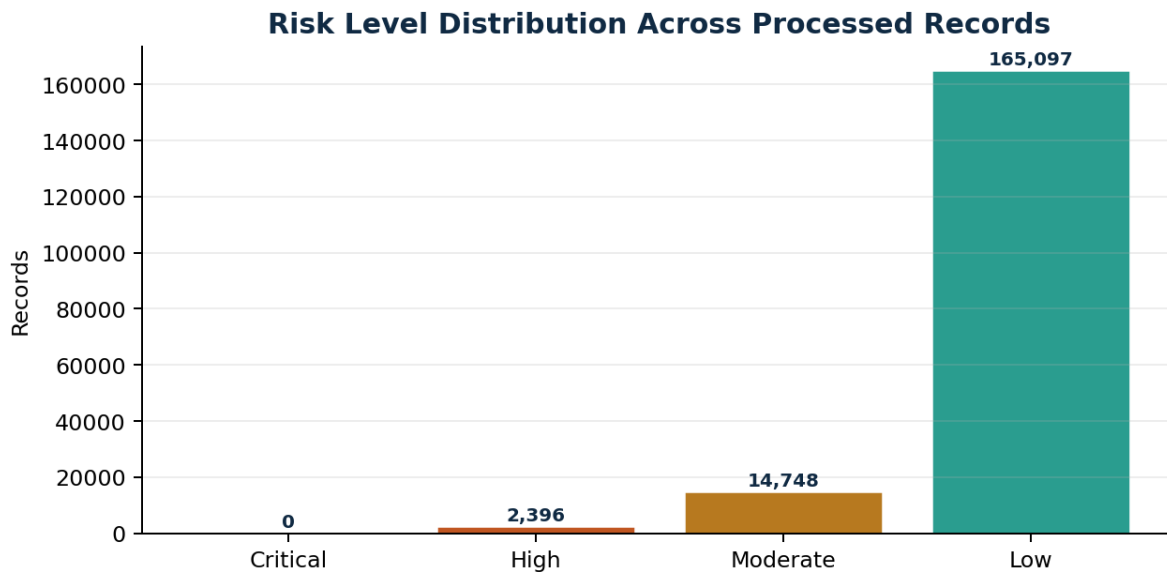


Figure 2. Risk-level distribution across the current processed surveillance dataset.

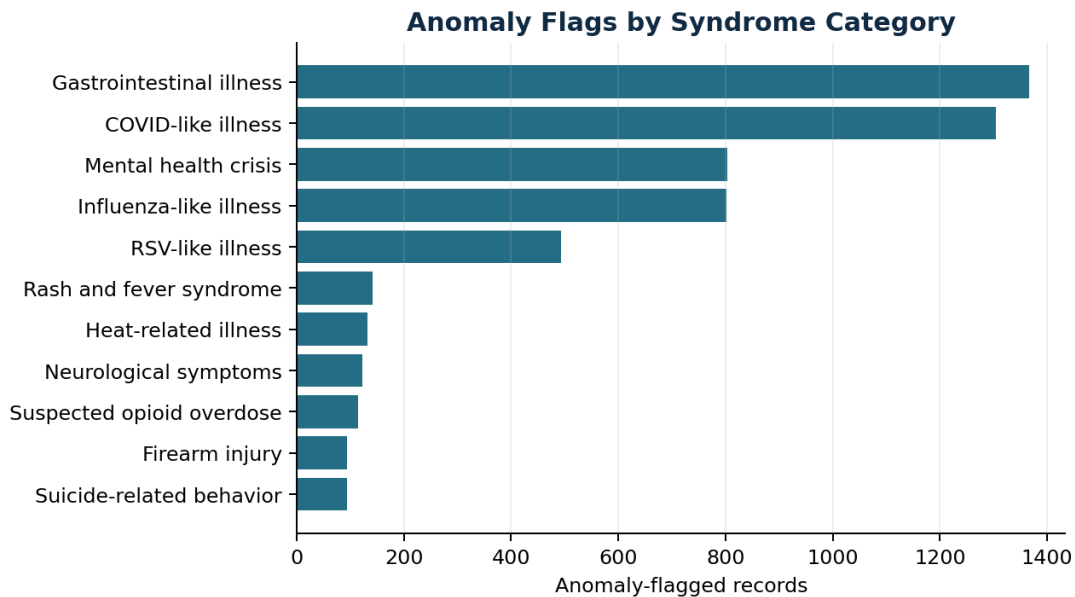


Figure 3. Anomaly flags by syndrome category.

## Forecasting Methodology

The forecasting module supports short-term trend review using ARIMA, Prophet when available, Exponential Smoothing fallback, and rolling-average fallback. Lightweight statistical forecasting is appropriate for the current prototype because it is interpretable, fast to run locally, and does not imply unsupported certainty.

Future work could evaluate LSTM or transformer-style sequence models if longer time histories, richer covariates, stable data feeds, labeled events, and model governance controls become available. For public health operations, forecast outputs should always include uncertainty language, validation metrics, and analyst review.

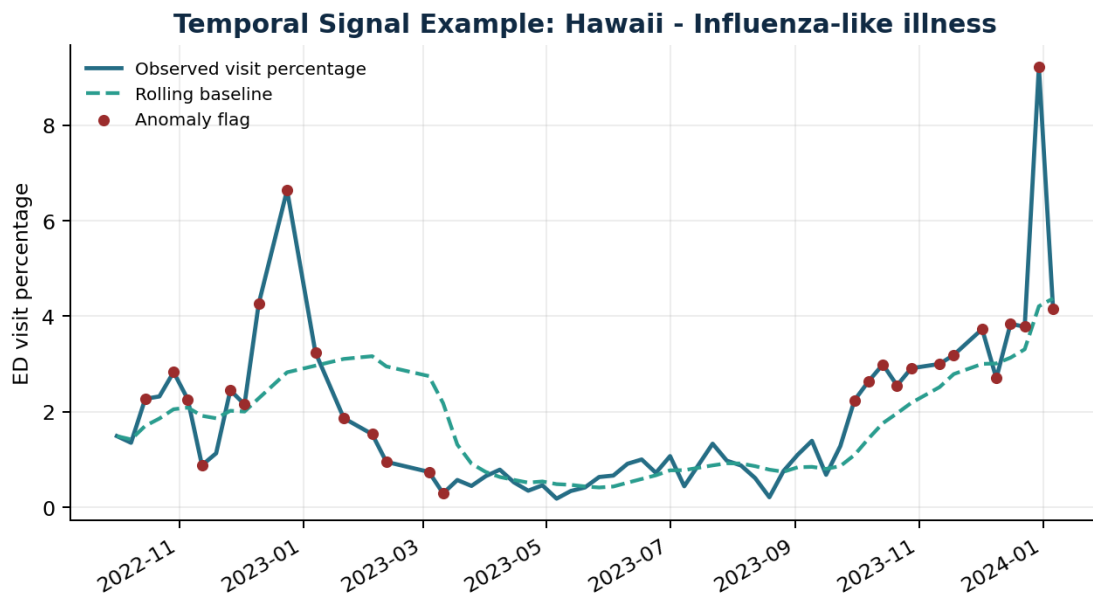


Figure 4. Example time-series signal with observed values, rolling baseline, and anomaly flags.

## RAG/NLP Surveillance Assistant

The retrieval-augmented assistant is designed to help analysts ask natural-language questions about surveillance trends, high-risk regions, anomaly explanations, data quality, syndrome definitions, methodology, and project documentation. It retrieves local documentation and a live surveillance snapshot before generating or formatting an answer.

This design reduces hallucination risk by grounding responses in local sources. When an LLM API key is not configured, the assistant falls back to retrieval-only answers. The assistant should not provide medical advice, diagnosis, treatment guidance, or official public health directives.

- Grounding sources include README, model summary, data dictionary, public health use cases, and live surveillance snapshot.
- Current retrieval uses TF-IDF and cosine similarity; future versions could use vector embeddings and ontology-aware retrieval.
- Governance controls should include audit logs, approved corpora, prompt boundaries, and refusal behavior for medical advice requests.

## Synthetic HL7 and Privacy-Preserving Simulation

The project includes synthetic HL7 ADT-style message generation for testing and demonstration. Synthetic messages can simulate emergency department visit metadata, chief complaints, and syndromic surveillance categories without exposing real patient information. This helps test ingestion concepts, classification logic, and training workflows.

Future synthetic data modules could use simulation or GAN-style methods to generate realistic but non-identifiable aggregate emergency department trend patterns. These synthetic patterns could stress-test anomaly detection and forecasting models under controlled scenarios such as outbreak spikes, reporting delays, seasonal surges, and data dropouts.

## Geospatial Risk Intelligence

Geospatial analytics translate model outputs into state- or region-level situational awareness. The dashboard aggregates latest risk scores by geography and displays rankings and map markers so analysts can identify where unusual activity may require closer review.

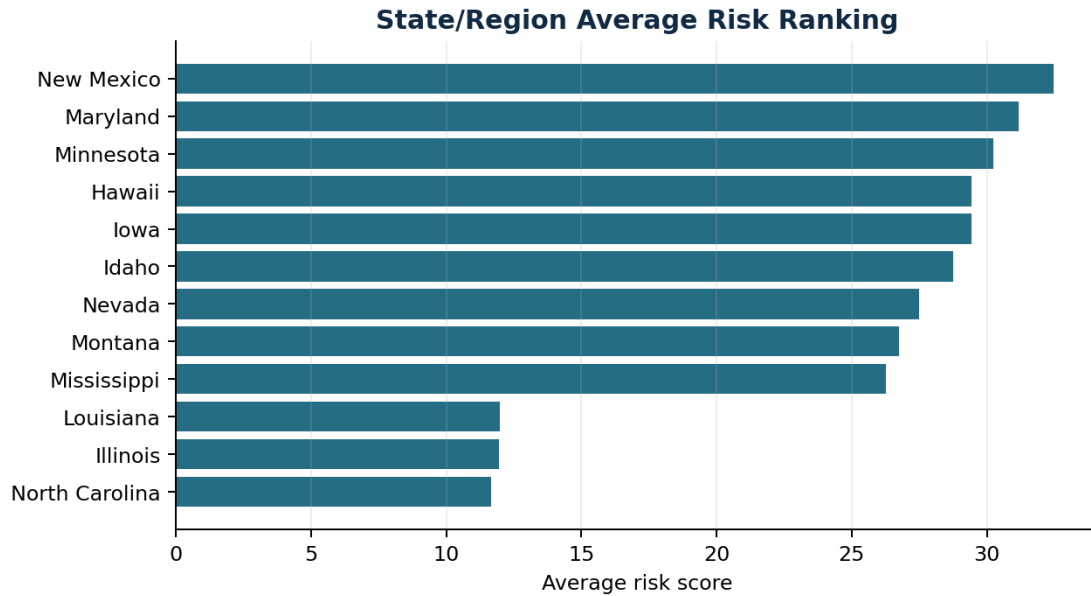


Figure 5. Average state/region risk ranking generated from processed model outputs.

| State / Region | Avg Risk | Max Risk | Anomalies | High/Critical Records |
|----------------|----------|----------|-----------|-----------------------|
| Hawaii         | 29.4     | 80.6     | 75        | 26                    |
| Iowa           | 29.4     | 80.4     | 77        | 36                    |
| Montana        | 26.8     | 79.9     | 69        | 35                    |
| Minnesota      | 30.2     | 79.8     | 78        | 47                    |
| New Mexico     | 32.5     | 79.5     | 87        | 43                    |
| Idaho          | 28.8     | 79.2     | 73        | 37                    |
| Mississippi    | 26.3     | 79.1     | 59        | 35                    |
| Louisiana      | 12.0     | 78.4     | 164       | 70                    |

## Quantitative Evaluation

The current dataset supports measurement of data coverage, anomaly prevalence, risk distribution, and data quality. Metrics such as precision, recall, sensitivity, and specificity require adjudicated ground truth labels and should not be asserted from unlabeled public or synthetic data alone.

| Metric                | Current Value | Interpretation                                                     |
|-----------------------|---------------|--------------------------------------------------------------------|
| Processed records     | 182,241       | Current analytic rows after cleaning and feature engineering.      |
| Anomaly flags         | 5,468         | Rows where z-score or Isolation Forest indicated unusual behavior. |
| Model agreement       | 0             | Rows where both anomaly detectors flagged the same record.         |
| Moderate records      | 14,748        | Rows mapped to Moderate risk.                                      |
| High/Critical records | 2,396         | Rows mapped to High or Critical risk.                              |

| Metric             | Current Value   | Interpretation                                        |
|--------------------|-----------------|-------------------------------------------------------|
| Average risk score | 12.50           | Mean composite risk score across all processed rows.  |
| Maximum risk score | 80.6            | Highest risk score in the current processed dataset.  |
| Completeness       | 99.58%          | Overall non-missing rate across processed fields.     |
| Precision/Recall   | Requires labels | Needs adjudicated events and analyst review outcomes. |

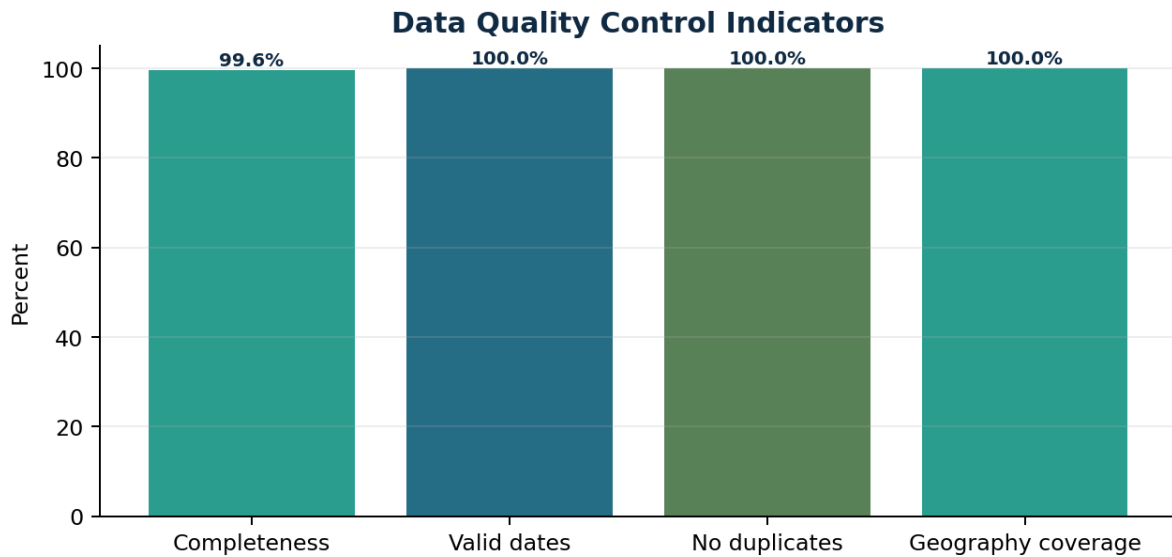


Figure 6. Data quality control indicators for the current processed dataset.

## Operational Readiness

Operational readiness requires more than functional models. A production system would need controls for security, privacy, auditability, data lineage, model governance, and human-in-the-loop review. Public health analytics should be designed to support expert judgment, not replace it.

| Domain           | Production Consideration                                                                                 |
|------------------|----------------------------------------------------------------------------------------------------------|
| Security         | Authentication, authorization, encryption, dependency scanning, secret management, and network controls. |
| Privacy          | Minimum necessary data, aggregation, de-identification, retention policy, and privacy impact review.     |
| Interoperability | HL7 v2, FHIR resources, vocabulary standards, and jurisdiction-specific syndrome definitions.            |
| Model governance | Versioning, threshold calibration, drift monitoring, validation, model cards, and change control.        |
| Auditability     | Input dataset version, scoring timestamp, model version, user activity, and alert disposition history.   |
| Responsible AI   | Human review, uncertainty communication, refusal of medical advice, and documented limitations.          |

## Future Roadmap

- Real-time or scheduled data refresh with monitored ingestion jobs.
- HL7/FHIR ingestion patterns and schema validation.
- County-level or jurisdiction-level geospatial views with privacy-aware aggregation.
- API-based alert publishing and analyst disposition workflow.
- Model monitoring, drift detection, threshold governance, and retrospective validation.



- Ontology-aware RAG using public health and clinical vocabularies.
- Privacy-preserving synthetic trend simulation for benchmark scenarios.
- Cloud-native deployment with role-based access, audit logging, and observability.

## Limitations

- The dataset is public aggregate or synthetic and should not be interpreted as live operational surveillance intelligence.
- The system is not affiliated with, endorsed by, or operated by any government agency.
- Model thresholds are prototype settings and require calibration before operational use.
- Forecasts are short-term analytic demonstrations, not official predictions.
- Precision and recall require adjudicated ground truth labels.
- The RAG assistant depends on retrieved local context and should be reviewed by a human analyst.
- The system must not be used for clinical decision-making, diagnosis, treatment, or operational public health response.

## Official References

CDC National Syndromic Surveillance Program: <https://www.cdc.gov/nssp/>

CDC BioSense Platform: <https://www.cdc.gov/nssp/php/about/about-nssp-and-the-biosense-platform.html>

CDC ESSENCE: <https://www.cdc.gov/nssp/php/onboarding-toolkits/essence.html>

CDC Data Modernization: <https://www.cdc.gov/data-modernization/php/about/index.html>

HHS Emergency Preparedness and Response: <https://www.hhs.gov/programs/emergency-preparedness/index.html>

WHO Epidemic Intelligence from Open Sources: <https://www.who.int/initiatives/eios>

## Disclaimer

This document describes an independent public health analytics research prototype using public aggregate or synthetic data. It is not an official system of HHS, CDC, WHO, Nssp, BioSense, ESSENCE, or any government agency. It is not intended for clinical decision-making, diagnosis, treatment, or operational public health response.